

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 6. Loops - Control Flow

Lecture: 62. Challenge - Prime Number

Section 1: Lecture Summary

This lecture focuses on solving two programming challenges using the Python for loop: (1) finding all factors of a given number, and (2) checking whether a given number is prime. The instructor explains the concept of factors as numbers that divide the given number exactly (with zero remainder). Using examples (like $n=15$ and $n=12$), the process of determining factors by testing divisibility from 1 up to n is demonstrated. The instructor then outlines and implements a simple Python program to print all factors of a number using a for loop and the modulo operator.

The next challenge is to determine if a number is prime, which means it has exactly two factors: 1 and itself. The approach builds on the factor-finding method but instead counts the number of factors. If the count is exactly two, the number is prime; otherwise, it is not. The instructor encourages students to write the prime-checking program themselves but provides the algorithm and sample code to guide them.

Section 2: Key Concepts and Explanations

- Factors of a Number:

- A factor of a number (n) is any integer (i) such that $(n \bmod i = 0)$ (there is no remainder).
- Factors are found by checking all numbers from 1 to (n) .
- The smallest factor is always 1, and the largest factor is (n) itself.
- Example: Factors of 15 are 1, 3, 5, and 15.

- Finding Factors Using a For Loop:

- Use `range(1, n+1)` to generate all numbers from 1 to n .
- For each number i , check if $n \bmod i == 0$.
- If true, then i is a factor.
- This approach has a simple and direct logic, suitable for beginner programmers.

- Prime Number Definition:

- A prime number is an integer greater than 1 that is divisible only by 1 and itself.
- Equivalently, a prime number has exactly two distinct factors.

- Checking for Prime Numbers:

- Use the factor-finding method to count the number of factors of n .
- Initialize a counter `count` to zero.
- For each i in `range(1, n+1)`, if $n \bmod i == 0$, increment `count`.
- After the loop, if `count == 2`, print "Prime"; otherwise, print "Not a prime".

Section 3: Example Code and Use Cases

Finding Factors of a Number:

```
n = int(input("Enter a number: "))
for i in range(1, n + 1):
    if n % i == 0:
        print(i)
```

Explanation: This code reads an integer `n` from the user, iterates from 1 to `n`, and prints all numbers `i` that divide `n` exactly (i.e., factors).

Checking if a Number is Prime:

```
n = int(input("Enter a number: "))
count = 0
for i in range(1, n + 1):
    if n % i == 0:
        count += 1
if count == 2:
    print("Prime")
else:
    print("Not a prime")
```

Explanation: This program counts the number of factors of `n`. If the count equals two, it confirms `n` is prime; otherwise, not.

Section 4: Key Takeaways

- Factors of a number are integers that divide the number without leaving a remainder.
- To find factors programmatically, iterate through all numbers from 1 to `n` and check divisibility using modulo `%`.
- A prime number is defined by having exactly two factors: 1 and itself.
- Counting factors during iteration helps determine if a number is prime.
- Both problems demonstrate effective use of the `for` loop and conditional statements in Python.
- The challenges reinforce understanding of loops, conditionals, modulo operation, and basic algorithmic thinking.

Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.



Challenges : for Loop

Challenges : for Loop

- Factors of a Number

Challenges : for Loop

- Factors of a Number
- Check if Prime Number

Factors of a Number Challenge

Factors of a Number
Challenge

n = 15

Factors of a Number
Challenge

n = 15

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
---	---	---	---	---	---	---	---	---	----	----	----	----	----	----

Factors of a Number

Challenge

n = 15

15 % 1 = 0

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

Factors of a Number

Challenge

n = 15

15 % 2 = 1

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

Factors of a Number

Challenge

n = 15

15 % 3 = 0

1 ~~2~~ 3 4 5 6 7 8 9 10 11 12 13 14 15

Factors of a Number

Challenge

n = 15

15 % 4 = 3

1 ~~2~~ 3 4 5 6 7 8 9 10 11 12 13 14 15

Factors of a Number
Challenge

n = 15

$$15 \% 4 = 3$$

1 ✖ 3 ✖ 5 6 7 8 9 10 11 12 13 14 15

Factors of a Number
Challenge

n = 15

$$15 \% 5 = 0$$

1 ✕ 3 ✕ 5 6 7 8 9 10 11 12 13 14 15

Factors of a Number
Challenge

n = 15

$$15 \% 6 = 3$$

1 ✕ 3 ✕ 5 6 7 8 9 10 11 12 13 14 15

Factors of a Number

Challenge

n = 15

15 % 7 = 1

1 ~~x~~ 3 ~~x~~ 5 ~~x~~ 7 8 9 10 11 12 13 14 15

Factors of a Number

Challenge

n = 15

15 % 8 = 7

1 ~~x~~ 3 ~~x~~ 5 ~~x~~ ~~x~~ ~~x~~ 9 10 11 12 13 14 15

Factors of a Number

Challenge

n = 15

15 % 9 = 6

1 ~~×~~ 3 ~~×~~ 5 ~~×~~ ~~×~~ ~~×~~ ~~×~~ 10 11 12 13 14 15

Factors of a Number

Challenge

n = 15

15 % 10 = 5

1 ~~×~~ 3 ~~×~~ 5 ~~×~~ ~~×~~ ~~×~~ ~~×~~ ~~×~~ 11 12 13 14 15

Factors of a Number

n = 15

15 % 11 = 4

1 ~~2~~ 3 ~~4~~ 5 ~~6~~ ~~7~~ ~~8~~ ~~9~~ ~~10~~ ~~11~~ 12 13 14 15

Factors of a Number

n = 15

$$15 \% 12 = 3$$

1 ~~2~~ 3 ~~4~~ 5 ~~6~~ ~~7~~ ~~8~~ ~~9~~ ~~10~~ ~~11~~ ~~12~~ 13 14 15

Factors of a Number

n = 15

15 % 13 = 2

1 ~~2~~ 3 ~~4~~ 5 ~~6~~ ~~7~~ ~~8~~ ~~9~~ ~~10~~ ~~11~~ ~~12~~ ~~13~~ 14 15

Factors of a Number

n = 15

15 % 14 = 1

1 ~~2~~ 3 ~~4~~ 5 ~~6~~ ~~7~~ ~~8~~ ~~9~~ ~~10~~ ~~11~~ ~~12~~ ~~13~~ ~~14~~ 15

Factors of a Number
Challenge

n = 15

$$15 \% 15 = 0$$

1 ✕ 3 ✕ 5 ✕ ✕ ✕ ✕ ✕ ✕ ✕ ✕ ✕ ✕ 15

Factors of a Number

Challenge

n = 15

1

☒

3

☒

5

☒

☒

☒

☒

☒

☒

☒

☒

☒

☒

15

Factors of a Number

Challenge

n = 15

1

3

5

15

Factors of a Number
Challenge

n = 15

factors = (1, 3, 5, 15)

Factors of a Number
Challenge

n = 12

1	2	3	4	5	6	7	8	9	10	11	12
---	---	---	---	---	---	---	---	---	----	----	----

Factors of a Number

Challenge

n = 12

12 %

12

1 2 3 4 5 6 7 8 9 10 11 12

Factors of a Number

Challenge

n = 12

12 % 1

1

1 2 3 4 5 6 7 8 9 10 11 12

Factors of a Number

Challenge

n = 12

12 % 1 = 0

1

2

3

4

5

6

7

8

9

10

11

12

Factors of a Number

Challenge

n = 12

12 % 2

1

2

3

4

5

6

7

8

9

10

11

12

Factors of a Number

Challenge

n = 12

12 % 2 = 0

1

2

3

4

5

6

7

8

9

10

11

12

Factors of a Number

Challenge

n = 12

12 % 3

1

2

3

4

5

6

7

8

9

10

11

12

Factors of a Number
Challenge

n = 12

$$12 \% 3 = 0$$

1	2	3	4	5	6	7	8	9	10	11	12
---	---	---	---	---	---	---	---	---	----	----	----

Factors of a Number
Challenge

$$n = 12$$

$$12 \% 4$$

1	2	3	4	5	6	7	8	9	10	11	12
---	---	---	---	---	---	---	---	---	----	----	----

Factors of a Number
Challenge

$$n = 12$$

$$12 \% 4 = 0$$

1	2	3	4	5	6	7	8	9	10	11	12
---	---	---	---	---	---	---	---	---	----	----	----

Factors of a Number
Challenge

$$n = 12$$

$$12 \% 5$$

1 2 3 4 5 6 7 8 9 10 11 12

Factors of a Number
Challenge

$$n = 12$$

$$12 \% 5 = 2$$

1 2 3 4 5 6 7 8 9 10 11 12

Factors of a Number

Challenge

n = 12

12 %

121110987654321

Factors of a Number

Challenge

n = 12

12 % 6

121110987654321

Factors of a Number

Challenge

n = 12

12 % 6 = 0

1

2

3

4

✗

6

7

8

9

10

11

12

Factors of a Number

Challenge

n = 12

12 % 7

1

2

3

4

✗

6

7

8

9

10

11

12

Factors of a Number

Challenge

n = 12

12 % 7 = 5

1

2

3

4

✗

6

7

8

9

10

11

12

Factors of a Number

Challenge

n = 12

12 % 7 = 5

1

2

3

4

✗

6

✗

8

9

10

11

12

Factors of a Number
Challenge

$$n = 12$$

$$12 \% 8 = 4$$

1	2	3	4	✗	6	✗	✗	9	10	11	12
---	---	---	---	---	---	---	---	---	----	----	----

Factors of a Number
Challenge

$$n = 12$$

$$12 \% 9 = 3$$

1 2 3 4 ✗ 6 ✗ ✗ ✗ 10 11 12

Factors of a Number
Challenge

$$n = 12$$

$$12 \% 10 = 2$$

1 2 3 4 ✗ 6 ✗ ✗ ✗ ✗ 11 12

Factors of a Number

Challenge

n = 12

12 % 11 = 1

1 2 3 4 ✕ 6 ✕ ✕ ✕ ✕ ✕ 12

Factors of a Number

Challenge

n = 12

12 % 12 = 0

1 2 3 4 ✕ 6 ✕ ✕ ✕ ✕ ✕ 12

Factors of a Number

Challenge

n = 12

12 % 12 = 0

1 2 3 4 ✕ 6 ✕ ✕ ✕ ✕ ✕ 12

Factors of a Number

Challenge

n = 12

12 % 12 = 0

1 2 3 4 6 12

Factors of a Number
Challenge

n = 12

1 2 3 4 6 12

Factors of a Number
Challenge

n = 12

factors = (1, 2, 3, 4, 6, 12)

Factors of a Number

Challenge

```
n = int(input('Enter a Number'))
```

Factors of a Number

Challenge

```
n = int(input('Enter a Number'))  
for i in range(1, n + 1):
```

Factors of a Number

Challenge

```
n = int(input('Enter a Number'))  
for i in range(1, n + 1):  
    if n % i == 0:
```

Factors of a Number

Challenge

```
n = int(input('Enter a Number'))  
for i in range(1, n + 1):  
    if n % i == 0:  
        print(i)
```

Check if Prime Number
Challenge

Check if Prime Number
Challenge

Check if Prime Number

Challenge

n = 7

Check if Prime Number

Challenge

n = 7

1	2	3	4	5	6	7
---	---	---	---	---	---	---

Check if Prime Number

Challenge

n = 7

1

2

3

4

5

6

7

Check if Prime Number

Challenge

n = 7

1

2

3

4

5

6

7

Check if Prime Number

Challenge

n = 7

1 2 3 4 5 6 7

Count = 2

Check if Prime Number

Challenge

n = 7

1 2 3 4 5 6 7

Count = 2

'Prime Number'

Check if Prime Number
Challenge

n = 8

Check if Prime Number

Challenge

n = 8

1

2

3

4

5

6

7

8

Check if Prime Number

Challenge

n = 8

1

2

3

4

5

6

7

8

Check if Prime Number

Challenge

n = 8

1

2

3

4

5

6

7

8

Check if Prime Number

Challenge

n = 8

1

2

3

4

5

6

7

8

Check if Prime Number

Challenge

n = 8

1

2

3

4

5

6

7

8

Check if Prime Number

Challenge

n = 8

1

2

3

4

5

6

7

8

Count = 4

Check if Prime Number

Challenge

n = 8

1 2 3 4 5 6 7 8

Count = 4

'Not a Prime'

Check if Prime Number

Challenge

n = int(input('Enter a Number'))

Check if Prime Number

Challenge

```
n = int(input('Enter a Number'))  
count = 0
```

Check if Prime Number

Challenge

```
n = int(input('Enter a Number'))  
count = 0  
for i in range(    ):
```

Check if Prime Number Challenge

```
n = int(input('Enter a Number'))  
count = 0  
for i in range(1,    ):
```

Check if Prime Number

Challenge

```
n = int(input('Enter a Number'))  
count = 0  
for i in range(1, n + 1):
```

Check if Prime Number

Challenge

```
n = int(input('Enter a Number'))  
count = 0  
for i in range(1, n + 1):  
    if n % i == 0:
```

Check if Prime Number

Challenge

```
n = int(input('Enter a Number'))
count = 0
for i in range(1, n + 1):
    if n % i == 0:
        count += 1
```

Check if Prime Number

Challenge

```
n = int(input('Enter a Number'))
count = 0
for i in range(1, n + 1):
    if n % i == 0:
        count += 1

if count == 2:
```


Check if Prime Number

Challenge

```
n = int(input('Enter a Number'))
count = 0
for i in range(1, n + 1):
    if n % i == 0:
        count += 1

if count == 2:
    print('Prime')
```

Check if Prime Number

Challenge

```
n = int(input('Enter a Number'))
count = 0
for i in range(1, n + 1):
    if n % i == 0:
        count += 1

if count == 2:
    print('Prime')
else:
```

Check if Prime Number

Challenge

```
n = int(input('Enter a Number'))
count = 0
for i in range(1, n + 1):
    if n % i == 0:
        count += 1

if count == 2:
    print('Prime')
else:
    print('Not a Prime')
```